Vincent Dong

vince.dong@gmail.com | 614-218-2455 | Philadelphia, PA | linkedin.com/in/vydong | github.com/vyund

Last updated June 2026

Research Interests

My research goal is to aid clinical decision making through Artificial Intelligence-driven tools. My research area is in the application of Machine Learning and Computer Vision algorithms for healthcare-related tasks, with an emphasis on integrating multi-modal data into interpretable models. I am passionate in developing automated tools that are based on clinical knowledge to improve their adoption and integration into the clinical workflow.

EDUCATION

University of Pennsylvania — **Philadelphia, PA**

Ph.D. in Bioengineering — *Aug 2022 - Present*

HHMI-NIBIB Interfaces Program in Biomedical Imaging and Informatics Sciences

Advisor: Dr. Andrew Maidment

Perelman School of Medicine — *Aug 2022 - Jan 2024*

Case Western Reserve University — **Cleveland, OH**

M.S. in Computing & Info Science — *GPA: 3.67/4.0* — *Aug 2020 - Jan 2022*

Concentration in Artificial Intelligence

Advisor: Dr. Anant Madabhushi

B.S. in Computer Science — *GPA: 3.45/4.0* — *Aug 2016 - May 2020*

RESEARCH & WORK EXPERIENCE

University of Pennsylvania X-ray Physics Laboratory — **Philadelphia, PA**

Grad. Research Assistant — *Jun 2021 – Aug 2021, Aug 2022 – Present*

- Assessed the ability of radiomic features to characterize background parenchymal enhancement and CAD-mark regions of interest for improving breast cancer risk stratification through classification experiments, odds ratio analyses, and race-based stratification to evaluate risk disparities across populations.
- Investigated the utility of explainable radiomic features for characterizing breast tissue density by proposing an explainable-AI feature selection method to identify highly predictive and influential features for each BI-RADS density grade.
- Explored patch-based deep learning models for denoising digital mammograms by using virtual clinical trials to investigate the trade-off between denoising strength and structural detail preservation via various loss functions.

Real Time Tomography LLC — **Villanova, PA**

Machine Learning Engineer — *Dec 2020 – Jul 2022*

- Trained a deep learning segmentation model with a low-memory footprint to segment and locate key anatomical regions within digital mammograms.
- Fit support vector machine classifier to identify digital mammograms imaging views of mediolateral oblique (MLO) and craniocaudal (CC) view digital mammograms.
- Conducted hyperparameter sweeps to optimize support vector machines for specimen container classification.

Center for Computational Imaging and Personalized Diagnostics — **Cleveland, OH**

Grad. Research Assistant — *Sep 2019 – Jul 2022*

- Extracted radiomic shape and texture features from non-human primate CT and PET scans to classify segmented tuberculosis lesions based on clinician-determined histopathology labels by using tSNE embeddings and random forest classifiers.
- Designed novel classification pipeline to detect tuberculosis bacilli from stained sputum slides by using an ensemble of patch-level deep convolutional neural networks and support vector machines.
- Evaluated performance of transfer learning techniques in predicting therapeutic response in diabetic retinopathy patients from ultrawide-field fluorescein angiography and optical coherence tomography scans by using gradient-weighted class activation maps.

PUBLICATIONS

[1] **Dong, V.**, Costa, A., Ehsan, S., Zeballos Torrez, C. R., Edmonds, C., Nayak, A., McCarthy, A. M., Maidment A. D. A., Barufaldi, B., "Utility of radiomic features as a complement to commercial AI CAD for BI-RADS 4 patient classification," *In preparation* (2026).

[2] Barufaldi, B., Vimieiro, R. B., **Dong, V.**, Tomic, H., Cao, Q., Vieira, M. C., Bakic, P. R., Eby, P. R., Zackrisson, S., McCarthy, A. M., and Maidment, A. D. A., "Lesion detectability and masking disparity assessment in breast tomosynthesis across diverse populations using in-silico imaging trials," *IEEE Transactions on Medical Imaging*, doi:10.1109/TMI.2026.3701599, *Early access* (2026).

[3] do Nascimento, G. A. B., **Dong, V.**, Costa, A. C., Cavalcante, G. J., Nguyen, A., do Rêgo, T. G., Malheiros, Y., Silva Filho, T. M., Zeballos Torrez, C. R., Gee, J. C., McCarthy, A. M., Maidment, A. D. A., and Barufaldi, B., "Toward a robust lesion detection model in breast DCE-MRI: adapting foundation models to high-risk women," *Medical Imaging 2026: Physics of Medical Imaging.* International Society for Optics and Photonics (SPIE) (2026), vol. 13924, 139242N. doi:10.1117/12.3085839.

[4] Cherezov, D.*, Via, Laura E.*, Khaleel, Z.*, Klein, E.*, Reiss, R.*, **Dong, V.#**, Borish, H. J.#, Yang, H.#, Mariello, P., Barry III, C., Flynn, J., Madabhushi, A., Xie, Y. L., "Radiomic phenotyping of tuberculosis lesion histopathology from computed tomography scans of non-human primates," *In preparation* (2026). *(*, # authors contributed equally)*

[5] **Dong, V.**, Mankowski, W., Silva Filho T. M., McCarthy A. M., Kontos, D., Maidment A. D. A., and Barufaldi, B., "Robust evaluation of tissue-specific radiomic features for classifying breast tissue density grades," *Journal of Medical Imaging* **12**(Suppl. 2), S22010, doi:10.1117/1.JMI.12.S2.S22010 (2025).

[6] **Dong, V.**, Barufaldi, B., Mankowski, W., Silva Filho T. M., McCarthy A. M., Kontos, D., and Maidment A. D. A., "Explainable radiomics to characterize breast density and tissue complexity: preliminary findings," *17th International workshop on Breast Imaging (IWBI2024).* International Society for Optics and Photonics (SPIE) (2024), vol. 13174, 131741L. doi:10.1117/12.3027032 (May 2024).

[7] **Dong, V.**, Sevgi, D. D., Sil Kar, S., Srivastava, S. K., Ehlers, J. P., and Madabhushi, A., "Evaluating the utility of deep learning for predicting therapeutic response in diabetic eye disease," *Frontiers in Ophthalmology* **2**, doi:10.3389/fopht.2022.85210 (Aug 2022).

[8] **Dong, V.**, Maidment, T. D., Borges, L. R., Hopkins, K., Kuo, J., Milani, A., Ringer, P. A., and Ng, S., "Automated multi-class segmentation of digital mammograms with deep convolutional neural networks," *16th International Workshop on Breast Imaging (IWBI2022).* International Society for Optics and Photonics (SPIE) (2022), vol. 12286, 122860M. doi:10.1117/12.262662 (Jul 2022).

[9] **Dong, V.**, Maidment, T. D., Borges, L. R., Barufaldi, B., Ng, S., and Maidment A. D. A., "Assessment of patch-based mammogram denoising methods using virtual clinical trials and deep learning: trade-off between denoising strength and preservation of structural details," in [*Medical Imaging 2022: Physics of Medical Imaging*], International Society for Optics and Photonics (SPIE) (2022), vol. 12031, 120311W. doi:10.1117/12.2612900 (Apr 2022).

[10] Sil Kar, S., Sevgi, D. D., **Dong, V.**, Srivastava, S. K., Madabhushi, A., and Ehlers, J. P., "Multi-Compartment Spatially-Derived Radiomics From Optical Coherence Tomography Predict Anti-VEGF Treatment Durability in Macular Edema Secondary to Retinal Vascular Disease: Preliminary Findings," *IEEE J Transl Eng Health Med* **9**, 1000113 (Jul 2021).

PRESENTATIONS

[1] **Dong, V.**, Costa, A., Ehsan, S., Zeballos Torrez, C. R., Edmonds, C., Nayak, A., McCarthy, A. M., Maidment A. D. A., Barufaldi, B., "Utility of radiomic features as a complement to commercial AI CAD for BI-RADS 4 patient classification," in [*Pendergrass Research Symposium 2026*], (May 2026).

[2] do Nascimento, G. A. B., **Dong, V.**, Costa, A. C., Cavalcante, G. J., Nguyen, A., do Rêgo, T. G., Malheiros, Y., Silva Filho, T. M., Zeballos Torrez, C. R., Gee, J. C., McCarthy, A. M., Maidment, A. D. A., and Barufaldi, B., "Toward a robust lesion detection model in breast DCE-MRI: adapting foundation models to high-risk women," in [*Basser Scientific Symposium 2026*], (May 2026)

[3] **Dong, V.**, Nguyen, A., Ehsan, S., Zeballos Torrez, C. R., McDonald, E. S., Maidment A. D. A., McCarthy, A. M., Barufaldi, B., "Characterizing malignant breast tumors on DCE-MRI using 3D textural features," in [*Pendergrass Research Symposium 2025*], (Jun 2025).

[4] **Dong, V.**, Barufaldi, B., Mankowski, W., Silva Filho, T. M., McCarthy, A. M., Kontos, D., Maidment A. D. A., "Explainable radiomics to characterize breast density and tissue complexity: preliminary findings," in [*International Workshop on Breast Imaging 2024*], (Jun 2024).

[5] **Dong, V.**, Barufaldi, B., Mankowski, W., Silva Filho, T. M., McCarthy, A. M., Kontos, D., Maidment A. D. A., "Explainable radiomics to characterize breast density and tissue complexity: preliminary findings," in [*Pendergrass Research Symposium 2024*], (Jun 2024).

[6] **Dong, V.**, Barufaldi, B., Mankowski, W., Silva Filho, T. M., McCarthy, A. M., Kontos, D., Maidment A. D. A., "Explainable radiomics to characterize breast density and tissue complexity: preliminary findings," in [*DVC AAPM Young Investigators Symposium 2024*], (Mar 2024).

[7] **Dong, V.**, Maidment, T. D., Borges, L. R., Hopkins, K., Kuo, J., Milani, A., Ringer, P. A., and Ng, S., "Automated multi-class segmentation of digital mammograms with deep convolutional neural networks," in [*International Workshop on Breast Imaging 2022*], (May 2022).

[8] **Dong, V.**, Maidment, T. D., Borges, L. R., Barufaldi, B., Ng, S., and Maidment A. D. A., "Assessment of patch-based mammogram denoising methods using virtual clinical trials and deep learning: trade-off between denoising strength and preservation of structural details," in [*Medical Imaging 2021: Physics of Medical Imaging*], International Society for Optics and Photonics, SPIE, (Apr 2022).

[9] Sil Kar, S., Sevgi, D. D., **Dong, V.**, Srivastava, S. K., Madabhushi, A., and Ehlers, J. P., "Multi-compartment oct-derived radiomics features to predict anti-vegf treatment durability for diabetic macular edema," *Investigative Ophthalmology & Visual Science* **62**, 3-3 (Aug 2021).

[10] **Dong, V.**, Sevgi, D. D., Sil Kar, S., Srivastava, S. K., Ehlers, J. P., and Madabhushi, A., "Evaluating the utility of deep learning using ultra-widefield fluorescein angiography for predicting need for anti-vegf therapy in diabetic eye disease," *Investigative Ophthalmology & Visual Science* **62**, 2114–2114 (Jun 2021).

SKILLS

Programming Languages: Python, MATLAB, C++, Java
Frameworks: PyTorch, TensorFlow, Keras, OpenCV, PyRadiomics, LibSVM, nltk, scikit-learn, NumPy, Pandas
Tools: Visual Studio Code, git, wandb, LaTeX, Tableau

RELEVANT COURSEWORK

Advanced Biomedical Imaging Applications, Biomedical Image Analysis, Fundamental Techniques of Medical Imaging, Physics of Medical/Molecular Imaging *[University of Pennsylvania]*

Cancer Biology, Mechanisms of Disease and Therapeutic Intervention, Clinical Anatomy, Immunology, Epidemiology, Reproduction, Cell & Tissue Biology, Endocrinology, Genetics *[Perelman School of Medicine]*

Computer Vision, Computational Perception, Statistical Natural Language Processing, Computational Intelligence, Probabilistic Graphical Models, Machine Learning, Software Engineering *[Case Western Reserve University]*